

Phase 5: EM Waves & Maxwell

The Unification of Electromagnetics

Master Study Guide & Step-by-Step Derivations

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1 Faraday's Law of Electromagnetic Induction

In electrostatics (Phase 1), we established that the electric field is strictly conservative. The curl of a static electric field is always zero ($\nabla \times \mathbf{E} = \mathbf{0}$). However, in 1831, Michael Faraday proved experimentally that this is *not* true when magnetic fields are allowed to change over time.

1.1 Derivation of the Differential Form (Highly Tested PYQ)

Problem Statement:

Derive the expression $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and show the pictorial view of this physical relation.

Step 1: The Macroscopic Experimental Law

Faraday discovered that if you push a magnet through a loop of wire, a voltage is induced in the wire. This induced voltage (Electromotive Force, or V_{emf}) is exactly equal to the negative time rate of change of the total magnetic flux (Φ) enclosed by that specific wire loop.

$$V_{\text{emf}} = -\frac{d\Phi}{dt} \quad (1)$$

The negative sign is incredibly important. It represents **Lenz's Law**, which dictates that nature abhors a change in flux. The induced voltage will attempt to drive a current in a direction that creates its *own* opposing magnetic field to fight the change.

Step 2: Defining EMF and Flux Mathematically

The electromotive force is fundamentally defined as the electrical work done to move a unit positive charge completely around the closed physical path L . Therefore, it can be written as the closed contour line integral of the electric field $\mathbf{E}$:

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} \quad (2)$$

Furthermore, the total macroscopic magnetic flux Φ is formally defined as the open surface integral of the magnetic flux density vector field $\mathbf{B}$ penetrating an arbitrary surface S bounded exactly by the contour loop L :

$$\Phi = \iint_S \mathbf{B} \cdot d\mathbf{S} \quad (3)$$

Step 3: Synthesizing the Integral Equation

We rigorously substitute Equation 2 and Equation 3 directly back into Faraday's macro-

scopic empirical law (Equation 1):

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \left(\iint_S \mathbf{B} \cdot d\mathbf{S} \right) \quad (4)$$

Step 4: Transformer EMF Assumption

Let us assume that the physical boundary loop L and its bounding area S are completely stationary in space. The loop is not physically moving, spinning, or vibrating. The only thing changing is the strength of the magnetic field itself. This is known as "Transformer EMF". Because the spatial surface S is constant, we can safely slide the full time derivative $\frac{d}{dt}$ inside the spatial surface integral. Since $\mathbf{B}$ is a multivariate function of both space and time, the total derivative transitions into a partial time derivative:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \iint_S \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{S} \quad (5)$$

This beautifully establishes the **Integral Form** of Faraday's Law.

Step 5: Applying Stokes' Theorem

To transition from a macro-scale integral equation to a microscopic point-form differential equation, we invoke Stokes' Theorem on the left-hand side. Stokes' Theorem mathematically converts the closed contour line integral of a vector field into an open surface integral of the *curl* of that vector field over the bounded area:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} \quad (6)$$

Substitute this geometric conversion back into Equation 5:

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = \iint_S \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{S} \quad (7)$$

Step 6: The Final Deduction

We algebraically group all terms together under a single unified surface integral:

$$\iint_S \left[\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \right] \cdot d\mathbf{S} = 0 \quad (8)$$

Because our initial physical choice of the bounded area S was completely arbitrary, this integral relationship must hold rigorously true for any conceivable surface, regardless of its size, shape, or orientation. The only way an integral can evaluate to zero for all possible domain boundaries is if the internal integrand itself is identically zero everywhere in space.

$$\begin{aligned}\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= \mathbf{0} \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}\end{aligned}\tag{9}$$

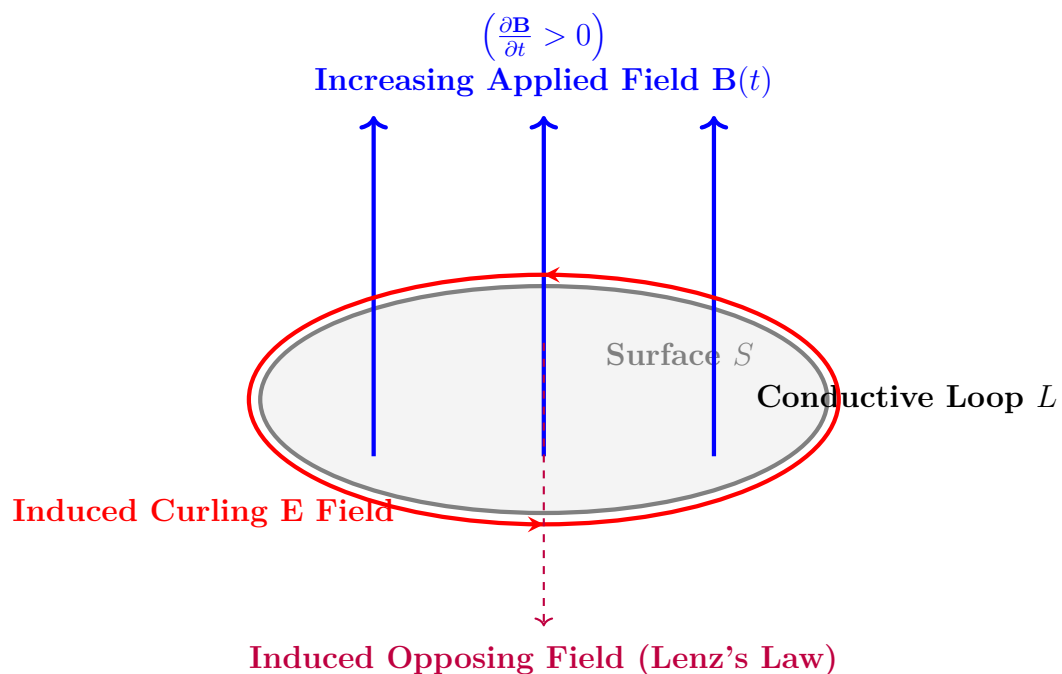
Final Result & Physical Conclusion:

The differential equation $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ proves that a time-varying magnetic field acts as a direct, fundamental physical source that generates a curling, non-conservative electric field in space.

1.2 Pictorial View of Faraday's Law

The university exams frequently ask for the "pictorial view" of this relationship to ensure you understand the spatial and temporal vectors involved.

TikZ Diagram: Faraday's Law and Lenz's Opposition



Explanation of the Pictorial View:

- The solid blue vectors represent the external, applied magnetic flux density $\mathbf{B}$. We assume this field is getting stronger (increasing) over time, meaning its derivative is positive.
- According to Faraday's Law, this changing field induces a circulating, non-conservative Electric field $\mathbf{E}$ around the boundary of the loop (shown in red).
- Following the right-hand rule and incorporating the negative sign from Lenz's Law, the induced $\mathbf{E}$ field drives a current clockwise.
- This clockwise current generates its *own* internal magnetic field (dashed purple vector) that points rigidly downwards, directly fighting and opposing the upward increase of the original external field.

2 The Displacement Current (Fixing Ampere's Law)

In Phase 4, we derived Ampere's Circuital Law ($\nabla \times \mathbf{H} = \mathbf{J}$). However, James Clerk Maxwell discovered that this law contains a fatal mathematical and physical flaw when fields are allowed to change over time.

2.1 Derivation of the Ampere-Maxwell Law (Highly Tested PYQ)

Problem Statement:

Establish $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ and explain the physical necessity of the displacement current.

Step 1: The Mathematical Contradiction

Let us assume that Ampere's static law is universally true:

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (10)$$

To stress-test this equation, we take the spatial divergence ($\nabla \cdot$) of both sides:

$$\nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \mathbf{J} \quad (11)$$

We know a fundamental vector calculus identity: the divergence of the curl of any vector field is always identically zero. Therefore, the left-hand side vanishes:

$$0 = \nabla \cdot \mathbf{J} \quad (12)$$

However, we must reconcile this result with the fundamental law of Charge Conservation, codified by the Continuity Equation:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \quad (13)$$

Comparing Equation 12 and Equation 13 reveals a catastrophic contradiction. Ampere's Law mathematically forces $\frac{\partial \rho_v}{\partial t} = 0$. This means Ampere's Law **only works for steady DC currents**. If you have an AC circuit or a discharging capacitor where charge density changes over time, Ampere's Law completely breaks down.

Step 2: Maxwell's Correction

To fix this fatal flaw, Maxwell hypothesized that an additional, unseen current density term must exist. Let us call it $\mathbf{J}_d$. He modified Ampere's Law to:

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d \quad (14)$$

Step 3: Solving for the Missing Term

We stress-test this new modified equation by taking the divergence again:

$$\nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot (\mathbf{J} + \mathbf{J}_d) \quad (15)$$

$$0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{J}_d \quad (16)$$

Rearranging to isolate the unknown Maxwell term:

$$\nabla \cdot \mathbf{J}_d = -\nabla \cdot \mathbf{J} \quad (17)$$

Substitute the true Continuity Equation (Equation 13) into this result:

$$\nabla \cdot \mathbf{J}_d = -\left(-\frac{\partial \rho_v}{\partial t}\right) = \frac{\partial \rho_v}{\partial t} \quad (18)$$

Step 4: Linking back to Electric Fields

We want to express this new current in terms of fields, not charges. We invoke the differential form of Gauss's Law ($\rho_v = \nabla \cdot \mathbf{D}$). Substitute this into Equation 18:

$$\nabla \cdot \mathbf{J}_d = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{D}) \quad (19)$$

Because space (x, y, z) and time (t) are independent mathematical variables in classical physics, we can safely swap the order of the spatial divergence operator and the temporal partial derivative:

$$\nabla \cdot \mathbf{J}_d = \nabla \cdot \left(\frac{\partial \mathbf{D}}{\partial t}\right) \quad (20)$$

By dropping the divergence operators on both sides, we identify the exact mathematical formulation for Maxwell's missing term:

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} \quad (21)$$

Final Result & Physical Conclusion:

Substituting this defined term back into the modified equation gives the universally valid **Ampere-Maxwell Law**:

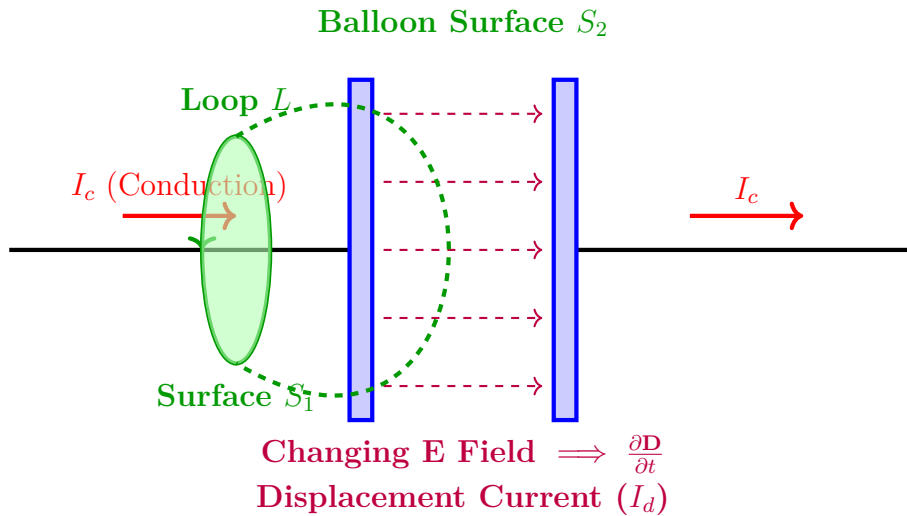
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

The term $\frac{\partial \mathbf{D}}{\partial t}$ is the **Displacement Current Density**. It proves that a changing electric field in empty space creates a magnetic field, just like a physical wire carrying electrons does.

2.2 Pictorial View: The Capacitor Paradox

To visually demonstrate why the displacement current is an absolute physical necessity, exams often require you to draw the "Capacitor Paradox" circuit.

TikZ Diagram: The Capacitor Paradox



Explanation of the Paradox: Imagine an AC circuit charging a capacitor. We draw a circular Amperian loop L around the left wire. Ampere's Law states: $\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$. We can calculate I_{enc} by integrating current density over *any* open surface bounded by L .

- **Scenario 1 (Surface S_1):** We choose a flat, taut disk stretching directly across the loop. The physical copper wire pierces directly through it. Therefore, $I_{\text{enc}} = I_c$ (Conduction current). The math works fine.

- **Scenario 2 (Surface S_2):** We choose to stretch the surface out like a balloon so that the membrane passes completely *between* the capacitor plates. No physical wire pierces this balloon surface. No physical electrons are flowing through the vacuum gap. Therefore, conduction current $I_c = 0$.

This creates a paradox: the line integral around the exact same loop L yields two different answers depending purely on our arbitrary choice of the bounded surface geometry.

Maxwell's Displacement Current fixes this perfectly. While $I_c = 0$ through the balloon surface S_2 , there is a massive, time-varying electric field building up between the plates. This changing flux density creates a Displacement Current $I_d = \iint (\frac{\partial \mathbf{D}}{\partial t}) \cdot d\mathbf{S}$. The total enclosed current is $I_c + I_d$, which remains continuous and identical regardless of the chosen surface geometry!

3 Power Flow and the Poynting Theorem

Now that electric and magnetic fields are coupled together via time-derivatives, they can transport energy through space. The Poynting Theorem is the universal statement of the conservation of energy for electromagnetic fields.

3.1 Derivation of the Poynting Vector (PYQ Focus)

Problem Statement:

Derive the Poynting Theorem and explain the physical significance of the Poynting Vector.

Step 1: The Vector Identity

We begin with the standard vector calculus identity for the divergence of a cross product of two vector fields, $\mathbf{E}$ and $\mathbf{H}$:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}) \quad (22)$$

Step 2: Substituting Maxwell's Curl Equations

We possess the two dynamic curl equations from Maxwell:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (23)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \quad (24)$$

(Note: We expanded $\mathbf{J} = \sigma \mathbf{E}$ using Ohm's Law for a conductive medium, and used the constitutive relations $\mathbf{B} = \mu \mathbf{H}$ and $\mathbf{D} = \epsilon \mathbf{E}$).

We meticulously substitute these into the right-hand side of Equation 22:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot \left(-\mu \frac{\partial \mathbf{H}}{\partial t} \right) - \mathbf{E} \cdot \left(\sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \right) \quad (25)$$

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mu \left(\mathbf{H} \cdot \frac{\partial \mathbf{H}}{\partial t} \right) - \sigma (\mathbf{E} \cdot \mathbf{E}) - \epsilon \left(\mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \right) \quad (26)$$

Step 3: Algebraic Simplification

We recognize a fundamental calculus chain-rule trick for vectors:

$$\frac{1}{2} \frac{\partial}{\partial t} (\mathbf{A} \cdot \mathbf{A}) = \mathbf{A} \cdot \frac{\partial \mathbf{A}}{\partial t} \quad (27)$$

We apply this trick to both the $\mathbf{E}$ and $\mathbf{H}$ time derivative terms in Equation 26. Also, we

recognize $\mathbf{E} \cdot \mathbf{E} = |\mathbf{E}|^2 = E^2$.

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\frac{\mu}{2} \frac{\partial}{\partial t}(H^2) - \sigma E^2 - \frac{\epsilon}{2} \frac{\partial}{\partial t}(E^2) \quad (28)$$

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\sigma E^2 - \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) \quad (29)$$

Equation 29 is the Differential (Point) form of the Poynting Theorem.

Step 4: Volumetric Integration

To understand the macroscopic power flow, we integrate this entire equation over a fixed spatial volume V :

$$\iiint_V \nabla \cdot (\mathbf{E} \times \mathbf{H}) dv = - \iiint_V \sigma E^2 dv - \iiint_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dv \quad (30)$$

Finally, we apply the Divergence Theorem to the left side to convert the volume integral of the divergence into a closed surface integral representing outward flux:

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = - \iiint_V \sigma E^2 dv - \frac{d}{dt} \iiint_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dv \quad (31)$$

Final Result & Physical Conclusion:

Physical Interpretation of the Poynting Theorem:

Equation 31 is a perfect statement of Energy Conservation.

- **Term 1 (Left):** Defines the **Poynting Vector** $\mathbf{P} = \mathbf{E} \times \mathbf{H}$ [Watts/m²]. The surface integral represents the total electromagnetic power flowing outwardly through the boundary surface.
- **Term 2 (Right):** Represents the total Ohmic/Joule heating loss (σE^2) dissipated as heat within the conductive volume.
- **Term 3 (Right):** Represents the time rate of decrease of the total stored electrostatic and magnetostatic energy dynamically held within the volume.

In plain English: "The power flowing out of a box equals the rate at which stored energy inside the box decreases, minus the power burned up as heat."

4 Electromagnetic Wave Equations

4.1 Derivation in Free Space (PYQ Focus)

Problem Statement:

Obtain E.M. wave equations in free space. (8 marks)

Step 1: Free Space Assumptions

We assume a completely ideal, source-free, non-conducting medium (perfect vacuum). There are no free charges ($\rho_v = 0$) and no free conduction currents ($\mathbf{J} = \mathbf{0}$, $\sigma = 0$).

The four Maxwell's Equations simplify dramatically:

$$\nabla \cdot \mathbf{E} = 0 \quad (32)$$

$$\nabla \cdot \mathbf{H} = 0 \quad (33)$$

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \quad (34)$$

$$\nabla \times \mathbf{H} = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (35)$$

Step 2: The Electric Field Wave Equation

We begin by taking the spatial curl ($\nabla \times$) of both sides of Faraday's Law (Equation 34):

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\mu_0 \frac{\partial \mathbf{H}}{\partial t} \right) \quad (36)$$

Because space and time are independent, we pull the constants and time derivative outside the curl operation on the right side:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \quad (37)$$

We now substitute Ampere's Law (Equation 35) directly into the right-hand side to completely eliminate the magnetic field $\mathbf{H}$:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \frac{\partial}{\partial t} \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \\ \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \end{aligned} \quad (38)$$

To resolve the complex left-hand side, we utilize the universally applicable vector calculus identity for the "curl of a curl":

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \quad (39)$$

By substituting this identity back into Equation 38:

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (40)$$

Recall our initial physical assumption established in Equation 32: because the vacuum medium is entirely source-free, $\nabla \cdot \mathbf{E} = 0$. The gradient of zero is zero, causing the first term to completely vanish:

$$\begin{aligned} 0 - \nabla^2 \mathbf{E} &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \\ \nabla^2 \mathbf{E} &= \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \end{aligned} \quad (41)$$

Step 3: The Magnetic Field Wave Equation

The derivation for the magnetic field analog perfectly mirrors the electric field. Take the curl of Ampere's Law:

$$\nabla \times (\nabla \times \mathbf{H}) = \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) \quad (42)$$

Substitute Faraday's Law to eliminate $\mathbf{E}$:

$$\nabla \times (\nabla \times \mathbf{H}) = \epsilon_0 \frac{\partial}{\partial t} \left(-\mu_0 \frac{\partial \mathbf{H}}{\partial t} \right) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2} \quad (43)$$

Apply the curl identity and set $\nabla \cdot \mathbf{H} = 0$:

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2} \quad (44)$$

Final Result & Physical Conclusion:

Both fields independently satisfy the exact same canonical 3D d'Alembert wave equation:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0 \quad \text{and} \quad \nabla^2 \mathbf{H} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0$$

This is the ultimate triumph of classical physics. It proves that electric and magnetic fields can detach from circuits and propagate infinitely through the empty vacuum of space as self-sustaining waves.

5 Analytical Solution of Plane Waves

5.1 What is a Plane Wave?

A **Uniform Plane Electromagnetic Wave** is a specific, highly idealized mathematical solution to the wave equations we just derived. It is physically defined by two fundamental characteristics:

1. **Constant Phase Surfaces:** The planes of constant phase are infinite, perfectly flat geometric planes that are strictly perpendicular to the direction of wave propagation.
2. **Uniformity:** The physical magnitude and spatial direction of both the electric field (**E**) and the magnetic field (**H**) are absolutely constant and uniform everywhere across any one of these infinite transverse planes at any given instant in time.

5.2 Analytical Solution (PYQ Focus)

Problem Statement:

Find an analytical solution of the wave equation of Magnetic Field (**H**) in free space considering it as a plane wave.

We begin with the derived 3D wave equation for the magnetic field:

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2} \quad (45)$$

Step 1: Enforce Plane Wave Geometry

Let us mathematically restrict the wave propagation to occur strictly and exclusively along the positive z -axis. Because the definition of a plane wave demands that the fields are entirely uniform across the transverse $x - y$ plane, the magnetic field cannot vary with respect to the spatial variables x or y . Therefore, all spatial derivatives with respect to x and y must identically evaluate to zero ($\frac{\partial}{\partial x} = 0, \frac{\partial}{\partial y} = 0$).

The 3D Laplacian operator collapses entirely into a simple 1D second spatial derivative:

$$\nabla^2 \mathbf{H} = \frac{\partial^2 \mathbf{H}}{\partial z^2} \quad (46)$$

This simplifies the complex 3D partial differential wave equation into a manageable 1D scalar wave equation. Assuming the magnetic field is linearly polarized along the

y-axis, meaning $\mathbf{H} = H_y(z, t)\hat{\mathbf{j}}$:

$$\frac{\partial^2 H_y}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 H_y}{\partial t^2} \quad (47)$$

Step 2: Defining the Wave Velocity

By mathematical analogy to standard classical wave mechanics (e.g., waves on a guitar string), any wave governed by the equation $\frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ propagates at a velocity v . Thus, the phase velocity c of the electromagnetic wave in vacuum is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s (The Speed of Light!)} \quad (48)$$

Step 3: Time-Harmonic Phasor Solution

To obtain the specific analytical solution, we assume the wave fluctuates sinusoidally with a fixed angular frequency ω . We express the field using complex phasor notation:

$$H_y(z, t) = \text{Re} \left\{ \tilde{H}_y(z) e^{j\omega t} \right\} \quad (49)$$

Substitute this phasor assumption directly back into the 1D differential equation. The double time derivative $\frac{\partial^2}{\partial t^2}$ simply pulls out a factor of $(j\omega)^2 = -\omega^2$:

$$\begin{aligned} \frac{d^2 \tilde{H}_y}{dz^2} &= \mu_0 \epsilon_0 (-\omega^2) \tilde{H}_y \\ \frac{d^2 \tilde{H}_y}{dz^2} + \omega^2 \mu_0 \epsilon_0 \tilde{H}_y &= 0 \end{aligned} \quad (50)$$

Let us define the **phase constant** (or wave number) β :

$$\beta = \omega \sqrt{\mu_0 \epsilon_0} = \frac{\omega}{c} \quad (51)$$

The differential equation now takes the classic form of a standard second-order linear ordinary differential equation (ODE):

$$\frac{d^2 \tilde{H}_y}{dz^2} + \beta^2 \tilde{H}_y = 0 \quad (52)$$

Step 4: Final Time-Domain Recovery

The general mathematical solution to this standard ODE is a linear combination of forward-traveling $(-j\beta z)$ and backward-traveling $(+j\beta z)$ complex exponential waves. Assuming the physical source exists at $z = -\infty$ and there are no reflective boundaries,

we discard the backward-traveling wave:

$$\tilde{H}_y(z) = H_0 e^{-j\beta z} \quad (53)$$

To retrieve the real, physical time-domain analytical solution, we multiply by the time harmonic $e^{j\omega t}$ and explicitly take the real part using Euler's formula:

$$\begin{aligned} H_y(z, t) &= \text{Re} \left\{ H_0 e^{-j\beta z} e^{j\omega t} \right\} \\ H_y(z, t) &= \text{Re} \left\{ H_0 e^{j(\omega t - \beta z)} \right\} \\ H_y(z, t) &= H_0 \cos(\omega t - \beta z) \end{aligned} \quad (54)$$

Final Result & Physical Conclusion:

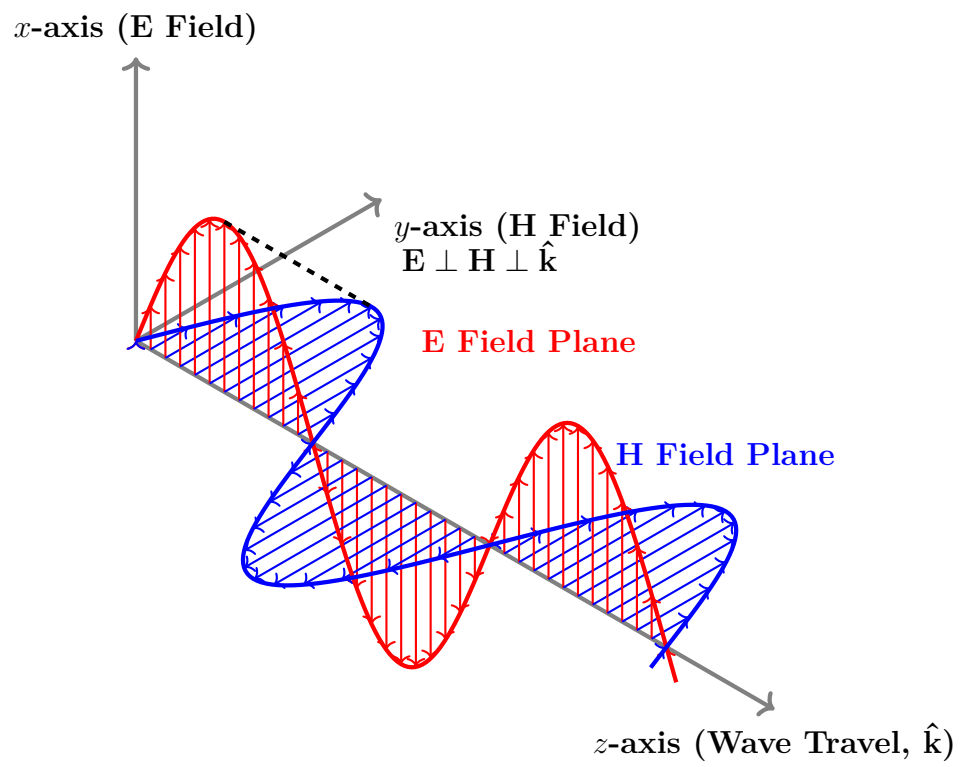
The fully analytical, time-domain solution representing the propagation of a plane magnetic wave traveling in the $+z$ direction in free space is:

$$\mathbf{H}(z, t) = H_0 \cos(\omega t - \beta z) \hat{\mathbf{j}} \quad [\text{A/m}]$$

5.3 Pictorial Drawing of EM Wave Propagation

The exam specifically asks you to draw the wave propagation. This is a Transverse Electromagnetic (TEM) wave, meaning both $\mathbf{E}$ and $\mathbf{H}$ are transverse (perpendicular) to the direction of travel, and perpendicular to each other.

TikZ Diagram: TEM Plane Wave Propagation



6 Phase 5 Summary and Final Exam Checklist

Congratulations! You have completed the grand unification of electromagnetic theory.

Before walking into your final exam, ensure you can confidently check off every item on this list:

- ☐ I can derive the differential form of Faraday's Law ($\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$) using Stokes' theorem, and I can draw the pictorial view showing Lenz's Law opposition.
- ☐ I can prove why Ampere's static law fails by showing that the divergence of the curl is zero, but the divergence of current density is not.
- ☐ I can derive the Displacement Current term ($\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$) using the Continuity Equation, and I can explain the Capacitor Paradox pictorial view.
- ☐ I can derive the Poynting Vector ($\mathbf{E} \times \mathbf{H}$) using the vector identity for $\nabla \cdot (\mathbf{E} \times \mathbf{H})$ and Maxwell's curl equations, and explain the three energy terms.
- ☐ I can derive the 3D Electromagnetic Wave equation for both $\mathbf{E}$ and $\mathbf{H}$ in free space by utilizing the "curl of a curl" identity ($\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$).
- ☐ I can collapse the 3D wave equation into a 1D scalar wave equation, solve it analytically using time-harmonic phasors to find $H_0 \cos(\omega t - \beta z)$, and draw the orthogonal TEM wave propagation.

You are now 100% prepared to dominate the Field Theory exam. Best of luck!